

ECON 282E — Homework 1, Part A

From a Question to a Quantitative Answer: Scope, Model, Validation, Data

Foundations of Macroeconomics
University of California, Riverside

Out: Thursday, October 8, 2026 • Due (with Part B): Thursday, October 29, 2026

The idea. This homework is a small research project, and you are its **project manager**. AI assistants can now write most of the code for you. What they cannot do for you is decide which question is worth asking, which model can answer it, what the equilibrium is, which method is good enough, which data make the model relevant, and whether the answer is right. Those decisions are the skill this homework trains, and they are what is graded.

The project. One model — the stochastic growth model of Sessions 1, 3, 4 and 5 — and two questions about the U.S. economy:

- **A positive question.** How much of U.S. business-cycle volatility can shocks to total factor productivity (TFP) account for?
- **A normative question.** What is the welfare cost of aggregate fluctuations, in consumption-equivalent terms — and why might the answer be very different in an economy where households cannot insure?

Part A (this sheet; Sessions 1–3) fixes the scope, writes the model down, validates your solver where the answer is known, and builds the data. **Part B** (out October 15; Sessions 4–5) chooses the algorithm, answers both questions, and puts it all together as a short paper, a five-slide deck and a replication package. **Both parts are due Thursday, October 29.**

What to hand in (with Part B). One PDF; a repository whose `run_all.py` reproduces every number and figure with one command; and an `AI_LOG.md`. Templates are posted beside this sheet (`homeworks-exams/HW1/templates/`).

AI policy. Assistance is allowed and expected. You remain responsible for every number, every citation and every line of code. **Validation is the part you cannot delegate.**

A0. The project charter (10 points)

Before writing any code, write a one-page charter (template `charter.md`). It must state:

- each question in one sentence, and whether it is positive or normative — and what makes it so;
- the model you will use, and **why a representative agent is enough for the positive question but only a benchmark for the normative one**. Use the complete-markets argument of Session 2 (full insurance, aggregation) to say exactly what the representative agent assumes away;
- the equilibrium object you will compute (a function? a number? a distribution?);
- the deliverables, and a **validation plan**: for each number you will report, the check that would convince a skeptical referee;

- (e) the three largest risks to the project (a method that is not accurate enough, data that do not measure what the model means, a calibration choice that drives the answer), and what you will do if each one happens.

You will revise the charter as you go; hand in the final version and say what changed.

A1. The model and its equilibrium, on paper (25 points)

A representative household has preferences $\mathbb{E}_0 \sum_{t \geq 0} \beta^t \ln c_t$ and supplies one unit of labour inelastically. A competitive firm produces $y_t = z_t k_t^\alpha n_t^{1-\alpha}$, capital depreciates at rate δ , and $\ln z_{t+1} = \rho \ln z_t + \sigma \varepsilon_{t+1}$ with $\varepsilon \sim N(0, 1)$.

- (a) **(6 pts)** Write the **social planner's** problem as a Bellman equation. State the state variables, the control and the feasible correspondence $\Gamma(k, z)$.
- (b) **(7 pts)** Define a **recursive competitive equilibrium** for the decentralized economy, in which households own capital and rent it and their labour to the firm. Say what the aggregate state is, which objects are *functions* of it, and what consistency condition links the household's and the aggregate law of motion. Then explain in three sentences why the welfare theorems let you compute this equilibrium by solving the planner's problem of (a).
- (c) **(6 pts)** For the deterministic version ($z_t \equiv Z$), verify Blackwell's sufficient conditions for the Bellman operator and say what, with the contraction mapping theorem, they let you conclude about existence, uniqueness and convergence.
- (d) **(6 pts)** **What the representative agent cannot say.** Now suppose each household also faces an uninsurable idiosyncratic income shock and can trade only a risk-free asset subject to $a' \geq -\underline{a}$ (the Aiyagari economy of Session 2). Without solving anything:
 - (i) define a stationary recursive competitive equilibrium for it;
 - (ii) say which step of the argument in (b) fails, and why the equilibrium interest rate satisfies $r^* < \rho_\beta \equiv 1/\beta - 1$;
 - (iii) say which of your two questions this matters for, and in which direction you expect it to move the answer.

A2. Validate the solver where the answer is known (30 points)

Before you trust a solver on a model with no closed form, test it on one that has one. Set $\delta = 1$, $z_t \equiv Z = 1$, $\alpha = 0.36$ and $\beta = 0.95$, so the problem is $\max \sum_t \beta^t \ln c_t$ subject to $c_t + k_{t+1} = Z k_t^\alpha$.

- (a) **(8 pts)** **The answer you will test against.** Guess $v(k) = A + B \ln k$ and verify it. Show that $g(k) = \alpha \beta Z k^\alpha$ and report A , B and the steady state k^* numerically.
- (b) **(7 pts)** **What “converged” means.** Using $\|v_n - v^*\| \leq \frac{\beta}{1-\beta} \|v_n - v_{n-1}\|$: if you stop when successive iterates differ by $\varepsilon = 10^{-6}$, how far from v^* might you still be at $\beta = 0.95$, and at $\beta = 0.99$ (the quarterly value you will meet in Part B)? How many sweeps does each need? What must a paper therefore report alongside “converged”?

- (c) **(8 pts) The unit test.** Solve by grid VFI on $[0.5k^*, 1.5k^*]$ with $N = 200$, tolerance 10^{-6} and float64, the choice restricted to the grid. Report the sweeps, $\max_k |v - v^*|$ and $\max_k |g/g^* - 1|$, and plot the computed policy against g^* over about 30 consecutive grid points. What does the plot show, and why does it not improve with more sweeps?
- (d) **(7 pts) How the error scales.** Solve at $N \in \{50, 100, 200, 400, 800\}$, and fit the slopes of $\log(\text{value error})$ and $\log(\text{policy error})$ against $\log h$. Compare with Santos & Vigo-Aguiar (1998). Then repeat at tolerance 10^{-11} : which slope changes, and why? Relate your answer to (b). *Write one sentence a referee would accept as your solver's error budget.*

A3. Make the model relevant: data and calibration (30 points)

- (a) **(12 pts) Structured data: the facts to be explained.** Download quarterly U.S. real GDP, real consumption, real investment and hours from FRED (Session 3, deck 3.B4, shows how). Record the **download date** (FRED revises data) and use 1954Q1–2019Q4 unless you argue for another sample. HP-filter the logs with $\lambda = 1600$ and report $\sigma(Y)$, $\sigma(C)/\sigma(Y)$, $\sigma(I)/\sigma(Y)$, $\text{corr}(C, Y)$, $\text{corr}(I, Y)$ and the first-order autocorrelation of Y . *This table is the target your model will be judged against in Part B; nobody hands it to you.*
- (b) **(10 pts) Calibrate.** Choose α , δ and β from long-run targets you can measure — for example the labour share, the depreciation rate of the capital stock, and the capital–output ratio — and state each target, its source series and its value. Then estimate ρ and σ for $\ln z$ from a Solow residual you construct (or from a published quarterly TFP series, such as Fernald's at the San Francisco Fed). Two choices will move your numbers: how you build capital, and how you detrend. **Make each choice, say why, and report how much the alternative would change ρ and σ .**
- (c) **(8 pts) Unstructured data, and an AI you must check.** Use an AI assistant to collect calibration values for α , β , δ , ρ and σ from at least three papers or official reports. Build a **calibration provenance table**: value, source, page or table, and whether *you* verified it in the original. Compare it with your own numbers from (b). In `AI_LOG.md`, record every value or citation the assistant got wrong and how you caught it. A table with an unverified entry marked as verified loses all 8 points.

A4. Red-team your AI assistant (5 points)

Ask an AI assistant for one derivation from this sheet — the recursive competitive equilibrium of A1(b), the Blackwell argument of A1(c), or the guess-and-verify of A2(a). Find a genuine error, gap or unstated assumption in its answer (a missing condition, a wrong sign, a step that does not follow, a result stated more generally than it holds). Quote the passage, explain what is wrong, and show the check that caught it — a counterexample, a numerical test, or the correct line of algebra. If you truly cannot find a flaw after pushing it with follow-up questions, document the three hardest challenges you posed and why its answers survived them.

Reporting standard, from here on. With every computed number: the method, the grid, the tolerance, the dtype, the hardware and the wall time; with every data number: the series ID, the sample and the download date. A number without these is not yet a result.